

Ritz Method — Gelfand - Fomin Problem 8.4

Austin Shank

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1 Problem Statement

4. Use the Ritz method to find an approximate solution of the problem of minimizing the functional

$$J[y] = \int_0^2 (y'^2 + y^2 + 2xy) dx, \quad y(0) = y(2) = 0,$$

and compare the answer with the exact solution.

Figure 1: Problem 8.4

2 Solution

2.1 Exact Solution

8.4	8/15/2026	
Problem: Find the extremal of $J[y] = \int_0^2 y'^2 + y^2 + 2xy \, dx$ $y(0) = y(2) = 0$ Soln: $F(x, y, y') = y'^2 + y^2 + 2xy$ $\left. \begin{aligned} F_y &= 2y + 2x \\ F_{y'} &= 2y' \\ \frac{d}{dx} F_{y'} &= 2y'' \end{aligned} \right\} \Rightarrow 2y + 2x - 2y'' = 0 \Rightarrow y'' - y = x$ Solve Euler's eqn complementary soln $y'' - y = 0 \Rightarrow m^2 - 1 = 0 \Rightarrow m = \pm 1 \Rightarrow y_c = C_1 e^x + C_2 e^{-x}$ particular soln $y_p = Ax + B \quad y_p' = A \quad y_p'' = 0$ substitute and solve A, B $-Ax - B = x \Rightarrow A = -1, B = 0 \Rightarrow y_p = -x$ general soln $y_g = C_1 e^x + C_2 e^{-x} - x$ Solve for C_1, C_2 $\left. \begin{aligned} y(0) = 0: \quad 0 &= C_1 + C_2 \Rightarrow C_2 = -C_1 \\ y(2) = 0: \quad 0 &= C_1 e^2 + C_2 e^{-2} - 2 \end{aligned} \right\} \Rightarrow 2 = C_1 e^2 - C_1 e^{-2} = C_1 (e^2 - e^{-2})$ $C_1 = \frac{2}{e^2 - e^{-2}} \quad C_2 = -\frac{2}{e^2 - e^{-2}}$ $y = \frac{2(e^x - e^{-x})}{e^2 - e^{-2}} - x = \frac{2 \sinh x}{\sinh 2} - x$ * We could have actually just used complementary soln of $y_c = C_1 \sinh(x) + C_2 \cosh(x)$, that would have probably been easier		

Figure 2: Problem 8.2

We will define the julia function of this for comparison against the approximations

```
y_exact(x) = 2 * sinh(x) / sinh(2) - x
have_exact = !isnothing(y_exact)
```

2.2 Ritz Approximations

We will use the RitzMethod package

```
using RitzMethod
import Optim
import Plots

# Define problem and basis
integrand(x, y, y_prime) = y_prime^2 + y^2 + 2 * x * y
domain = Interval(0.0, 2.0)
full_basis = [x -> x^k * (2 - x) for k in 1:3]

function solve_ritz(n_terms)
    minimizefunctional(
        integrand, domain, full_basis[1:n_terms];
        num_quad_nodes=5,
        method=Optim.BFGS(),
        options=Optim.Options(iterations=50, g_abstol=1e-20, x_reltol=1e-8),
    )
end

results = [solve_ritz(n) for n in 1:3]
```

2.2.1 One-Term Approximation

```
println(summarize(results[1]))
```

```
RitzResult:
  Domain:      [0.0, 2.0]
  Coefficients: [-0.35714285714285726]
  Minimized value: -0.47619047619047633
  Converged:    true
```

2.2.2 Two-Term Approximation

```
println(summarize(results[2]))
```

```
RitzResult:
  Domain:      [0.0, 2.0]
  Coefficients: [-0.20496894409937896, -0.15217391304347833]
  Minimized value: -0.5167701863354041
  Converged:    true
```

2.2.3 Three-Term Approximation

```
println(summarize(results[3]))
```

```

RitzResult:
  Domain:      [0.0, 2.0]
  Coefficients: [-0.22826086956521813, -0.09510869565217232, -0.02853260869565301]
  Minimized value: -0.5173913043478264
  Converged:    true

```

3 Comparison

Term count	Minimized J[y]	Coefficients
1	-0.47619048	[-0.35714285714285726]
2	-0.51677019	[-0.20496894409937896, -0.15217391304347833]
3	-0.51739130	[-0.22826086956521813, -0.09510869565217232, -0.02853260869565301]

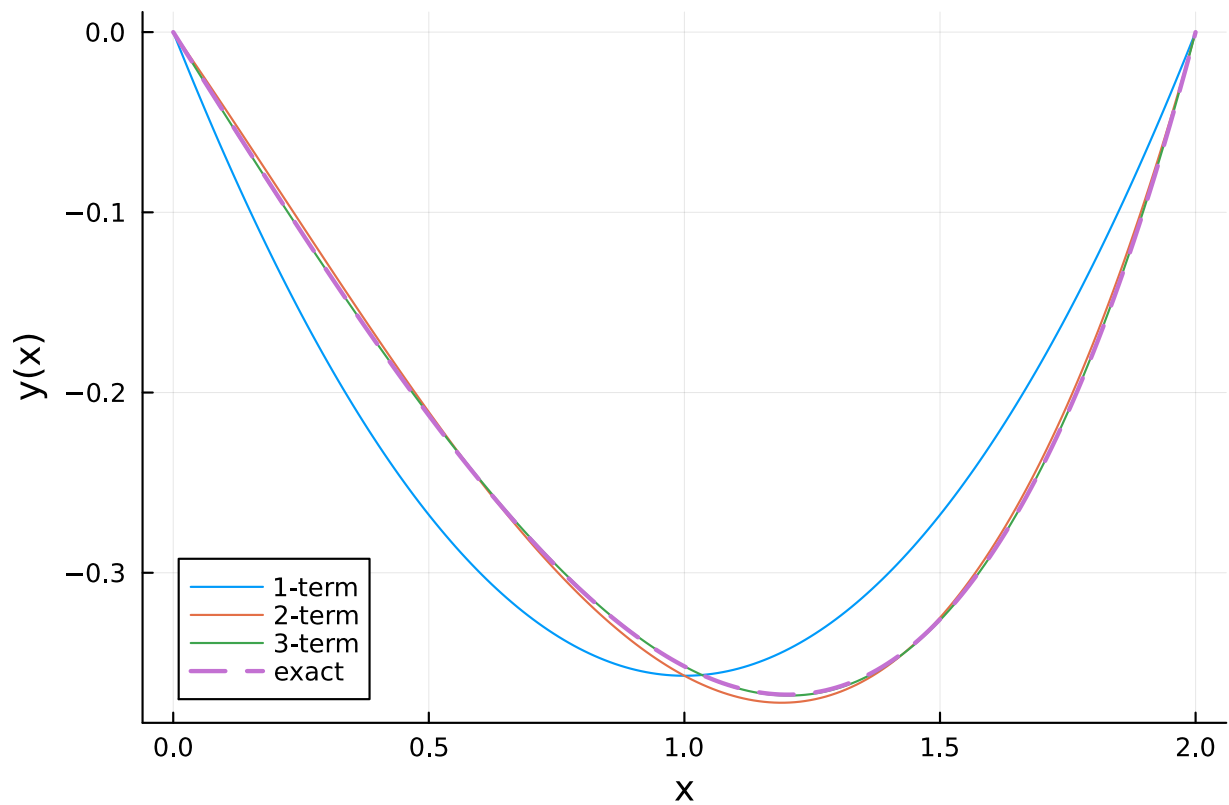


Figure 3: Ritz Approximations

4 Approximation Error

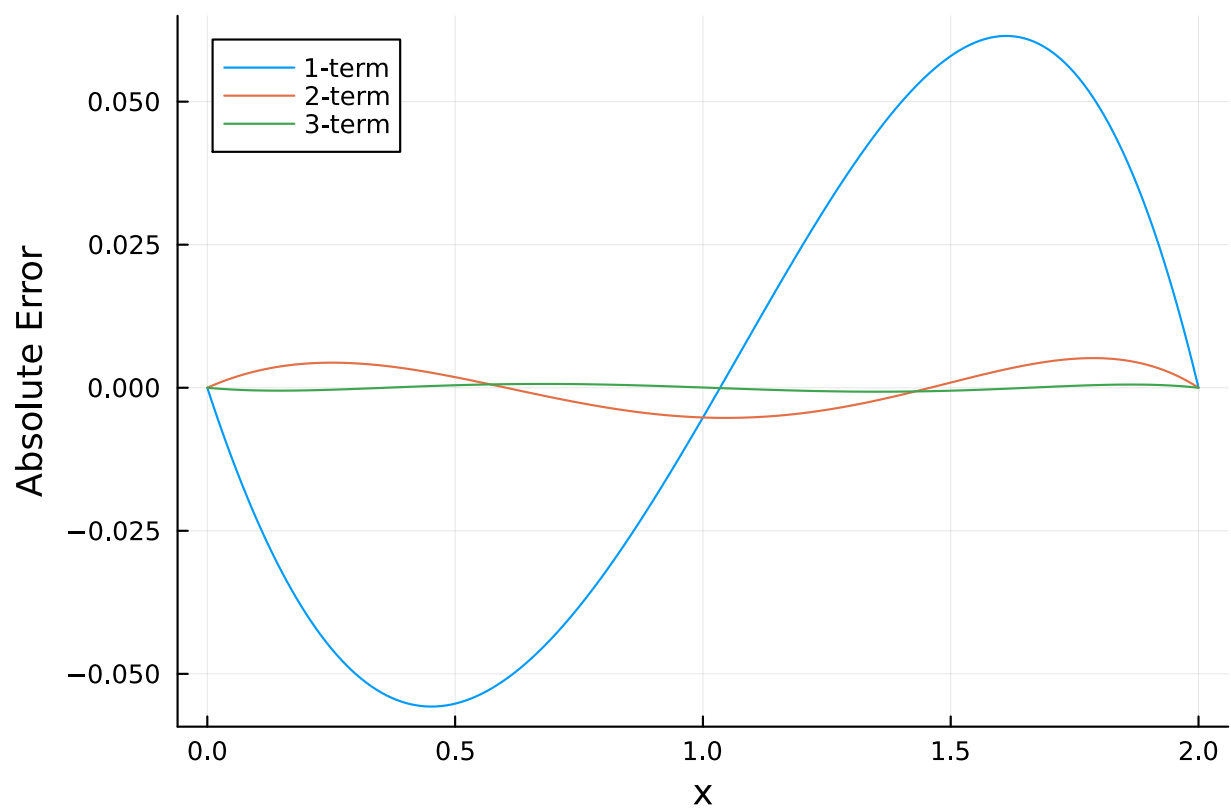


Figure 4: Pointwise error between Ritz approximations and the exact solution.